ECON6074: Causal Inference

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About the Course

This course surveys modern methods of applied microeconometric research. The aim of this course is to help students to apply the econometrics theory with data in the real world. This course will be empirically oriented, with an emphasis on causal identification using various methods such as randomized control trial, instrumental variable method, differences-in-differences, and regression discontinuity design. This course will focus on applied econometric issues with cross section and panel data.

Recommended Textbook

Angrist, Joshua D. and Jörn-Steffen Pischke. *Mostly Harmless Econometrics*. Princeton,
New Jersey: Princeton University Press, 2009.

Angrist, Joshua D. and Jörn-Steffen Pischke. *Mastering 'Metrics: The Path from Cause to Effect*. Princeton, New Jersey: Princeton University Press, 2015.

Evaluations

Empirical Assignments	40%
Paper Presentation	20%
Quiz	30%
Participation and Class Engagement	10%

(1) **Empirical Assignment.** (40%) Students are required to form groups of five and work on two (out of four) data assignments throughout the semester. Given different datasets, you will be asked to investigate causal effects and replicate empirical papers.

(2) **Group presentation of an empirical paper.** (20%)
We intend to have an active discussion of research papers -- and develop students' presentation skills. Students are required to form groups of five and take turns in presenting the assigned research papers. Each group will present one paper from the reading list below (papers without suffix * and #). The target length of presentation is

20-25 mins. The structure of presentation should at least include:

- Research questions
- Literature reviews
- Contributions
- Research design/Methodology (***)
- Empirical results
- Conclusion: any surprise? What do you learn from the paper?

(3) **Participation and Class Engagement.** (10%)
Courses will be offered face to face, and no recording will be arranged. You should come to each class. According the MEcon Programme policy, 70% of class attendance will be required. The base point is 0. Students will be awarded 2 points for each question they ask or answer. The TA will take attendance before lecture starting, and record the engagement. Students are expected to have completed the required readings before coming to class, to contribute to these discussions.

Course Outline & Readings
(* Required Readings, # Recommended Readings)

Topic 0: Introduction and Review

Chapter 1 & 2, Angrist, Joshua D. and Jörn-Steffen Pischke. *Mostly Harmless Econometrics*. Princeton, New Jersey: Princeton University Press, 2009. (*)

Topic I: Causal Inference, Randomized Trial and (Semi-)Natural Experiment

Chapter 3, Angrist, Joshua D. and Jörn-Steffen Pischke. *Mostly Harmless Econometrics*. Princeton, New Jersey: Princeton University Press, 2009. (*)

Jacob, Brian A., Max Kapustin, and Jens Ludwig. 2015. "The Impact of Housing Assistance on Child Outcomes: Evidence From A Randomized Housing Lottery." *Quarterly Journal of Economics* 130(1): 465-506. (*)

Bursztyn, Leonardo, Thomas Fujiwara, and Amanda Pallais. 2017. "'Acting Wife': Marriage Market Incentives and Labor Market Investments." *American Economic Review* 107(11): 3288-3319. (*)

Cantoni, Davide, David Y. Yang, Noam Yuchtman, and Y. Jane Zhang. 2019. "Protests as Strategic Games: Experimental Evidence from Hong Kong's Antiauthoritarian Movement." *Quarterly Journal of Economics* 134(2): 1021-1077.

Chen, Yuyu, and David Y. Yang. 2019. "The Impact of Media Censorship: 1984 or Brave New World." *American Economic Review* 109(6): 2294-2332.

Chodorow-Reich, Gabriel, Gita Gopinath, Prachi Mishra, and Abhinav Narayanan. 2019. "Cash and the Economy: Evidence from India's Demonetization." *Quarterly Journal of*

Economics 135(1): 57-103.

Cai, Jing, and Adam Szeidl. 2018. "Interfirm Relations and Business Performance." *Quarterly Journal of Economics* 133(3): 1229-1282.

Durante, Ruben, Paolo Pinotti, and Andrea Tesei. 2019. "The Political Legacy of Entertainment TV." *American Economic Review* 109(7): 2497-2530.

Topic II: IV/2SLS

Chapter 4, Angrist, Joshua D. and Jörn-Steffen Pischke. *Mostly Harmless Econometrics*. Princeton, New Jersey: Princeton University Press, 2009. (*)

Angrist, Joshua D., and Alan B. Krueger. 1991. "Does Compulsory School Attendance Affect Schooling and Earnings?" *Quarterly Journal of Economics* 106(4): 979-1014. (*)

Acemoglu, Daron, Simon Johnson, and James Robinson. 2001. "Colonial Origins of Comparative Development: An Empirical Investigation." *American Economic Review* 91(5): 1369-1401. (*)

Madestam, Andreas, Daniel Shoag, Stan Veuger, and David Yanagizawa-Drott. 2013. "Do Political Protest Matter? Evidence From The Tea Party Movement." *Quarterly Journal of Economics* 128(4): 1633-1685. (*)

Gilchrist, Duncan Sheppard, and Emily Glassberg Sands. 2016. "Something to Talk About: Social Spillovers in Movie Consumption." *Journal of Political Economy* 124(5): 1339-1382.

Pascali, Luigi. 2016. "The Wind of Change: Maritime Technology, Trade and Economic Development." *American Economic Review* 107(9): 2821-2854.

Schelnker, Wolfram, and W. Reed Walker. 2015. "Airports, Air Pollution, and Contemporaneous Health. " *Review of Economic Studies* 83(2): 768-809

Moser, Petra, Alessandra Voena, and Fabian Waldinger. 2014. "German Jewish Émigrés and US Invention." *American Economic Review* 104(10): 3222-3255.

Allcott, Hunt, Luca Braghieri, Sarah Eichmeyer, and Matthew Gentzkow. 2020. "The Welfare Effects of Social Media." *American Economic Review* 110(3): 629-676.

Topic III: Panel Data and Fixed Effects

Chapter 5, Angrist, Joshua D. and Jörn-Steffen Pischke. *Mostly Harmless Econometrics*. Princeton, New Jersey: Princeton University Press, 2009. (*)

Bertrand, Marianne, and Antoinette Schoar. 2003. "Managing with Style: The Effect of Managers on Firm Policies." *Quarterly Journal of Economics* 118(4): 1169-1208. (#)

Agarwal, Sumit, David Lucca, Amit Seru, and Francesco Trebbi. 2014. "Inconsistent Regulators: Evidence from Banking." *Quarterly Journal of Economics* 129(2): 889-938.

Dell, Melissa, Benjamin F. Jones, and Benjamin A. Olken. 2012. "Temperature Shocks and Economic Growth: Evidence from the Last Half Century." *American Economic Journal: Macroeconomics* 4(3): 66-95.

Storeygard, Adam. 2016. "Farther on Down the Road: Transport Costs, Trade and Urban Growth in Sub-Saharan Africa." *Review of Economic Studies* 83: 1263-1295

Hodler, Roland, and Paul A. Raschky. 2014. "Regional Favoritism." *Quarterly Journal of Economics* 129(2): 995-1033.

Topic IV: Differences-in-Differences

Chapter 5, Angrist, Joshua D. and Jörn-Steffen Pischke. *Mostly Harmless Econometrics*. Princeton, New Jersey: Princeton University Press, 2009. (*)

Bertrand, Marianne, Esther Duflo and Sendhil Mullainathan. 2004. "How Much Should We Trust Differences-in-Differences Estimates?" *Quarterly Journal of Economics* 119(1): 249-275. (#)

Card, David, and Alan B. Krueger. 1994. "Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania." *American Economic Review* 84(4): 772-793. (*)

Draca, Mirko, Stephen Machin, and Robert Witt. 2011. "Panic on the Street of London: Police, Crime, and the July 2005 Terror Attacks." *American Economic Review* 101(5): 2157-2181. (*)

Kearney, Melissa S., and Phillip B. Levine. 2015. "Media Influences on Social Outcomes: The Impact of MTV's 16 and Pregnant on Teen Childbearing." *American Economic Review* 105(12): 3597-3632. (*)

Agarwal, Sumit, and Wenlan Qian. 2014. "Consumption and Debt Response to Unanticipated Income Shocks: Evidence from a Natural Experiment in Singapore." *American Economic Review* 104(12): 4205-4230.

Bloom, Nicholas, Mirko Draca, and John Van Reenen. 2016. "Trade Induced Technical Change? The Impact of Chinese Imports on Innovation, IT and Productivity." *Review of Economic Studies* 83(1): 87-117.

Giroud, Xavier. 2013. "Proximity and Investment: Evidence From Plant-Level Data." *Quarterly Journal of Economics* 128(2): 861-915.

Pierce, Justin R., and Peter K. Schott. 2016. "The Surprisingly Swift Decline of US

Manufacturing Employment." *American Economic Review* 106(7): 1632-1662.

Adda, Jerome. 2016. "Economic Activity and the Spread of Viral Disease: Evidence from High Frequency Data." *Quarterly Journal of Economics* 131(2): 891-941.

Greenstone, Michael, Richard Hornbeck, and Enrico Moretti. 2010. "Identifying Agglomeration Spillovers: Evidence from Winners and Losers of Large Plant Openings" *Journal of Political Economy* 118(3): 536-598.

Depetris-Chauvin, Emilio, Ruben Durante, and Filipe Campante. 2020. "Building Nations Through Shared Experiences: Evidence from African Football." *American Economic Review* 110(5): 1572-1602.

Patrick, Kline, Neviana Petkova, Heidi Williams, and Owen Zidar. 2019. "Who Profits From Patents? Rent-Sharing at Innovative Firms." *Quarterly Journal of Economics* 134(3): 1343-1404.

Topic V: Regression Discontinuity

Chapter 6, Angrist, Joshua D. and Jörn-Steffen Pischke. *Mostly Harmless Econometrics*. Princeton, New Jersey: Princeton University Press, 2009. (*)

Lee, David S., Thomas Lemieux. 2010. "Regression Discontinuity Designs in Economics." *Journal of Economic Literature* 48(2): 281-355. (*)

Lee, David S. 2008. "Randomized Experiments From Non-Random Selection in U.S. House Elections." *Journal of Econometrics* 142(2): 675-97. (#)

Pop-Eleches, Cristian, and Miguel Urquiola. (2013). "Going to a Better School: Effects and Behavioral Responses." *American Economic Review* 103(4): 1289-1324. (*)

Crost, Benjamin, Joseph Felter, and Patrick Johnston. 2014. "Aid Under Fire: Development Projects and Civil Conflict." *American Economic Review* 104(6): 1833-1856.

Fujiwara, Thomas. 2015. "Voting Technology, Political Responsiveness, and Infant Health: Evidence from Brazil." *Econometrica* 83(2): 423-464.

Campante, Filipe, and David Yanagizawa-Drott. 2017. "Long-Range Growth: Economic Development in the Global Network of Air Links." *Quarterly Journal of Economics* 133(3):1395-1458.

Anderson, Siwan. 2018. "Legal Origins and Female HIV." *American Economic Review* 108(6):1407-1439.

Asher, Sam, and Paul Novosad. 2020. "Rural Roads and Local Economic Development." *American Economic Review* 110(3):797-823.

Ehrlich, Maximilian V., and Tobias Seidel. 2018. “The Persistent Effects of Place-Based Policy: Evidence from the West-German Zonenrandgebiet.” *American Economic Journal: Economic Policy* 10(4):344-374.